

Responsible AI

ECE 750T

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Department of Electrical and Computer Engineering

University of Waterloo

0 Course information

Territorial acknowledgement



We acknowledge that we live and work on the traditional territory of the **Neutral, Anishinaabeg, and Haudenosaunee** peoples.

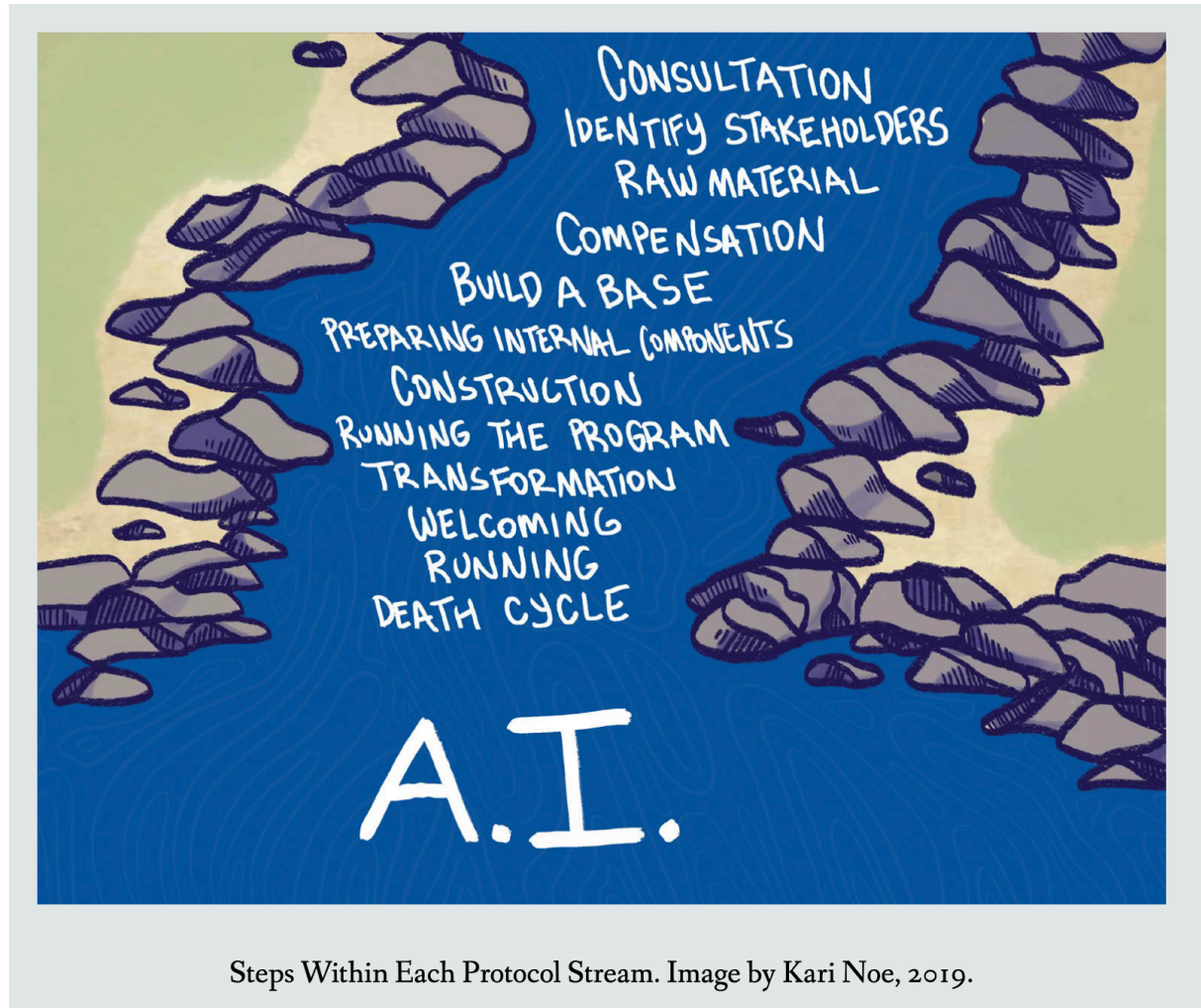
The University of Waterloo is situated on the **Haldimand Tract**, the land promised to the Six Nations that includes six miles on each side of the Grand River.

Indigenous Protocol and Artificial Intelligence

How to Build Anything Ethically

Suzanne Kite et al.

Indigenous Protocol and
Artificial Intelligence Position
Paper (2020)



whoami



Elliot Creager
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Hobbies: Guitar 🎸, birdwatching 🐦

Education: 🎓

- Ph.D., Computer Science
@ University of Toronto
- M.A., Music Technology
@ McGill University
- Sc.B., Electrical Engineering; A.B., Music
@ Brown University

Employment: 💻

- Research Intern
@ Google Brain
- Research Scientist
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whoarewe — Ambreesh



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Hobbies: Music production 🎵, football ⚽,
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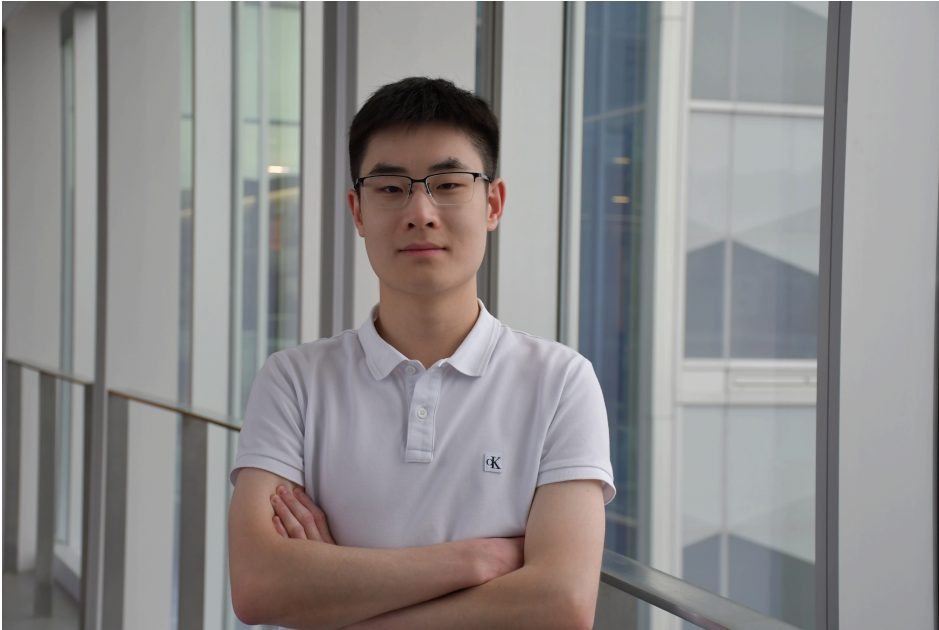
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- M.A.Sc., Electrical & Computer Engineering
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@ Shiv Nadar University

Employment: 💻

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- Analyst (Intern)
@ McKinsey & Co

whoarewe — Chenghao



Chenghao Tan
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Education: 🎓

- M.A.Sc., Electrical & Computer Engineering
@ University of Waterloo
- B.Eng., Computer Science
@ Hangzhou Dianzi University

Hobbies:

- Movies 🎬
- Traveling 🌍
- Open-source DIY 🔧

Who are you?

- Graduate students with a foundation in machine learning
- Different research interests and experiences
- A shared interest in building AI responsibly

What program are you in, and what experience do you bring? 🤔

Answer in class.

Where are we?

Meeting	Time	Location
Lecture	Fridays, 1:00–3:50pm	PSE 4053
Instructor office hours	By appointment	EIT 3116

- **LEARN:** announcements and course materials.
- **Piazza:** course questions and finding a presentation group; link forthcoming on LEARN.
- **Email the instructor:** personal matters, accommodations, illness, or missed work.
- Lectures **will not be recorded.**

What will we learn?

Understand the risks

- Fairness and measurement
- Evaluation and auditing
- Memorization and privacy
- Foundation-model safety

Assess possible responses

- Fair learning and private learning
- Data attribution and unlearning
- Data poisoning and collective action
- Governance and policy

Read research critically, lead a discussion, and carry out an **individual research project**.

What would you like to learn?

What does “Responsible AI” mean to you? 🤔

- Answer in class.

Which AI risks or questions would you like us to address? 🤔

- Answer in class.

How class will work

- **Weeks 1–2:** prof lectures and class discussion.
- **Weeks 3–10:** prof lecture, break, three student-led paper sessions.
 - student presentations in **groups of three**. sign-up sheet TBA.
 - you will present **once** this term. think about which week/topic you want to explore!
- **Week 6:** midterm exam replaces prof lecture.
- **Week 12:** individual research-project presentations.
- **Week ??:** guest lectures?

Assessment at a glance

Component	Weight	Work submitted by
Paper presentation & discussion	15%	Group (3 students) — shared mark
Micro-survey	5%	Group (3 students) — shared mark
Assignment 1: Fairness	10%	Each student
Assignment 2: Documentation audit	10%	Each student
Midterm exam	10%	Each student
Research project	50%	Each student

Project breakdown: proposal 5% · report 35% · presentation 10%.

Group presentation outline



1 · Present

Presenting group

Explain the paper and pose a question worth discussing.

2 · Reflect

Audience pairs

Think, then compare ideas with a neighbour.

3 · Discuss

Whole class

Group facilitates; classmates question, respond, and build on ideas.

4 · Synthesize

Presenting group

Draw together key insights and open questions; transition.

Your group leads → your group scribes. Divide responsibilities within the group.

Focus notes on **discussion and synthesis**; your presentation slides will also be shared.

Send the professor **Markdown notes** (write in Markdown or convert before sending). Notes are shared with the class.

Lead a discussion worth having

Shared 15% mark: 10% presentation + 5% facilitation.

- Explain the important ideas and evidence clearly.
- Ask accessible questions and follow up on classmates' reasoning.
- Connect contributions and synthesize what the discussion revealed.
- **Your group also scribes.** Capture the discussion; slides and notes will be shared with the class.

Post-discussion: writing a micro-survey

5%, shared group mark · Due Nov 27

- Compare your presented paper with **two others from the same week's reading list**.
- Connect their questions, assumptions, approaches, and findings.
- Incorporate an insight or question from the class discussion.
- Include a brief **author contribution statement**.

Assignments and midterm

Task	Out	Due / exam
A1: Fairness — 10%	Sep 18	Oct 2
A2: Documentation audit — 10%	Oct 3	Oct 19
Midterm exam — 10%	—	Oct 23, in class

- **A1:** fair classification, metrics, an intervention, and reflection.
- **A2:** audit dataset/model documentation and the validity of a measurement; no programming.
- **Midterm:** Weeks 1–5; conceptual questions and worked problems.

Your individual research project

50% of the course grade. All components are individual.

Component	Weight	Deliverable
Proposal — Week 5	5%	1–2 pages: question, motivation, plan
Report — end of term	35%	6–8 pages, excluding references
Presentation — Week 12	10%	Explain your research and findings

Choose a research question connected to the course and develop an appropriate methodology.

Submissions and deadlines

- Unless otherwise instructed, submit written work to **both Crowdmark and LEARN Dropbox**.
- Written submissions are due at **11:59pm** on the stated date.
- **Late submissions will not be accepted**, except where an accommodation applies.
- Micro-surveys: one submission per group.

See the course outline and assignment instructions for details.

Course policies: generative AI

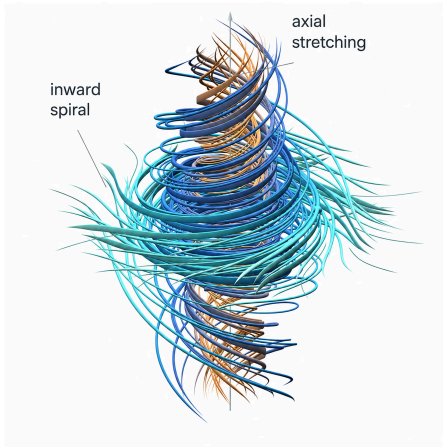
Generative AI tools produce text, images, music, or code—for example, ChatGPT, DALL-E, or GitHub Copilot. See the course outline for the full policy. 📌

- **GenAI may be used with proper documentation, citation, and acknowledgment ([how to cite](#)).**
- GenAI may falsify references, fabricate facts, or inaccurately express ideas.
- Content generated from other authors' work may contain inaccuracies or reflect biases.
- The legal and copyright status of GenAI inputs and outputs is unclear; see the [Copyright Advisory Committee](#).
- **You are accountable for the content and accuracy of all submitted work, including work supported by GenAI.**
- **Clarity matters:** Submitting “sloppy” work may result in poor marks.
 - Be prepared to reproduce any part of their submitted work for the instructor upon request

1 AI Risks and Taxonomies

What's New?

The Navier–Stokes controversy: AI, credit, and trust



For most of the past year progress was slow. We worked through the literature and upgraded various preliminary results, up to obtaining finite time blow up for the Incompressible Porous Media equation (with smooth forcing). This was until about a month ago, when we had real progress: on August 15th, we obtained the blow up results, with smooth forcing, for both Boussinesq and Euler. I can say the first LLM generated proof Levent sent me was the most horrendous I have ever read; we verified it on Lean on August 22nd. Since this point, we have been working around the clock to understand this proof and turn it into something readable.

There is another part of this story, and one that, honestly, I very much wish I did not have to be concerned with. I am not happy about the presentation quality in these papers. Ideally, we would have preferred to spend weeks turning the LLM generated proofs into something readable from the very first page.

Verification · Readability · Credit

How should researchers disclose AI use, and how should credit for new discoveries be assigned? 🤔

11 Sept. 2026 · Visualization: OpenAI (blog). Annotated statement: Buckmaster (2026), p. 1.

What are we asking AI to do?

Perceive

Identify an object in an image.

Predict

Estimate an outcome from records.

Generate

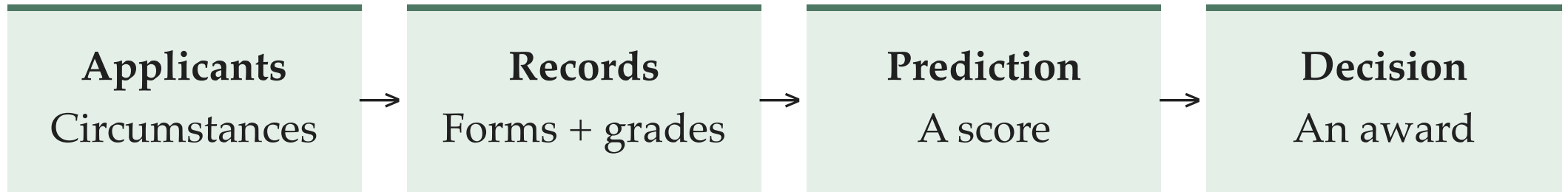
Draft text, images, or code.

Control

Choose actions as a situation unfolds.

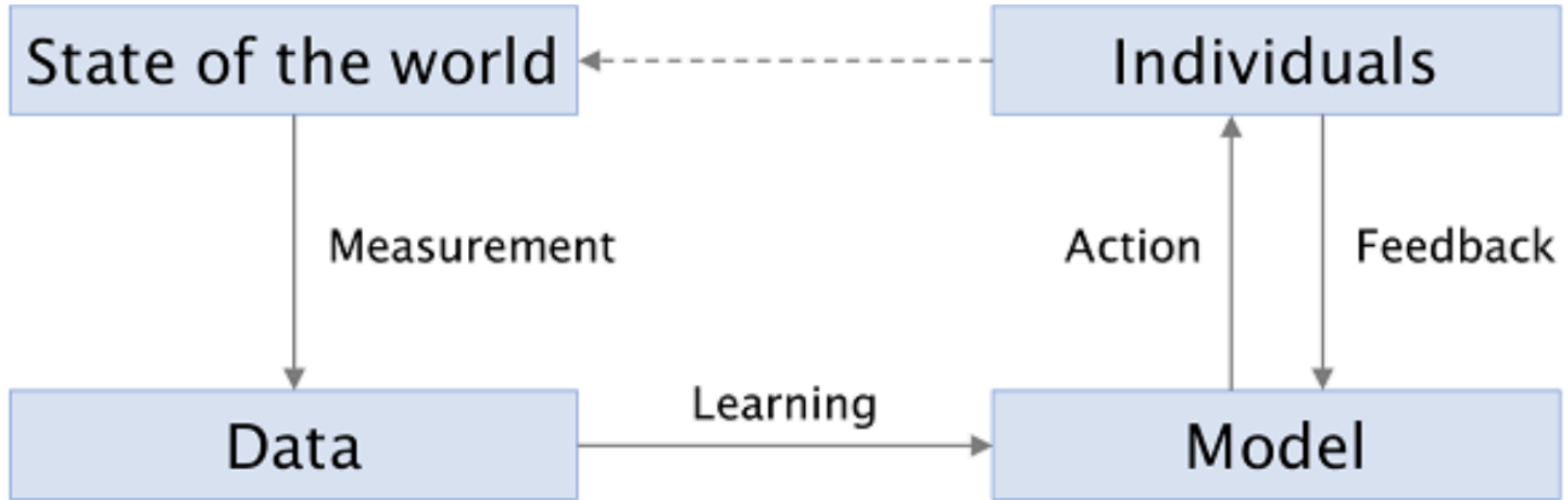
A scholarship service

Our hypothetical case: a university has a limited scholarship budget.



What should a scholarship help us achieve? 🤔

The machine learning loop



The model sits inside a larger system.

Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 1, p. 5.

What counts as “success”?

Goal: help students flourish

Future grades?

Academic performance

Financial need?

Access to resources

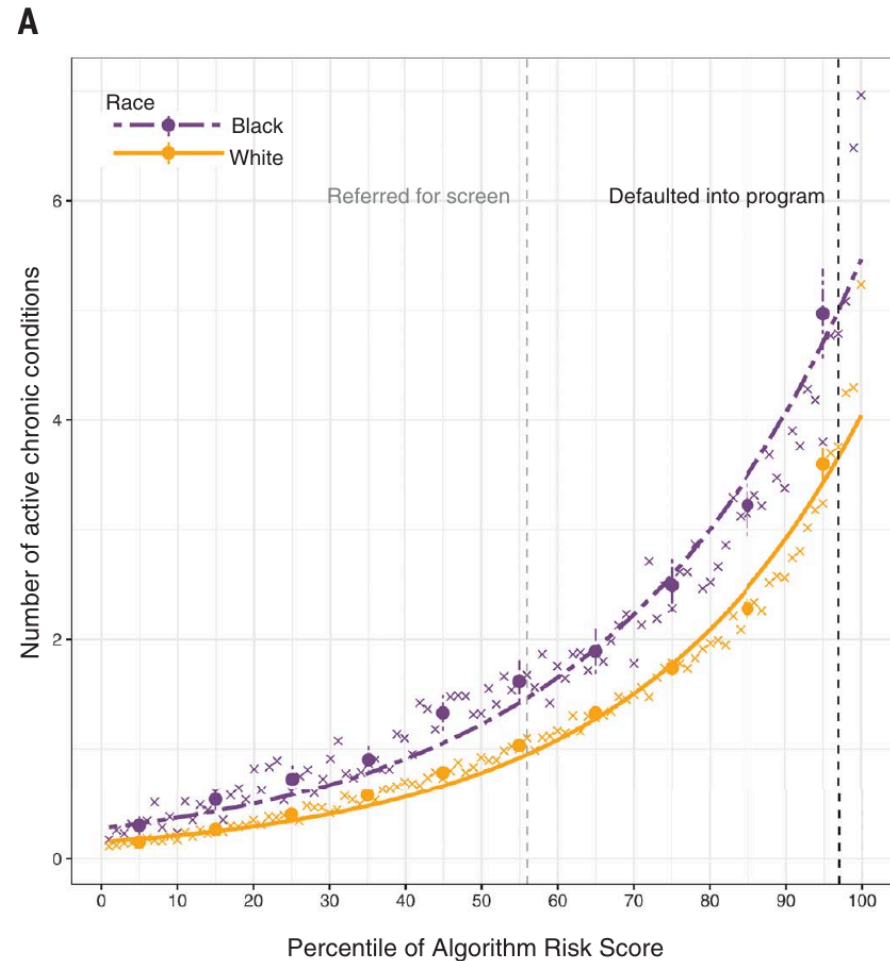
Past awards?

Previous decisions

What would each target leave out? 🤔

Hypothetical application of Barocas, Hardt, and Narayanan (2023), Chapter 1, “Measurement,” pp. 6–9.

When cost stands in for need



Intended use

Identify patients needing extra care.

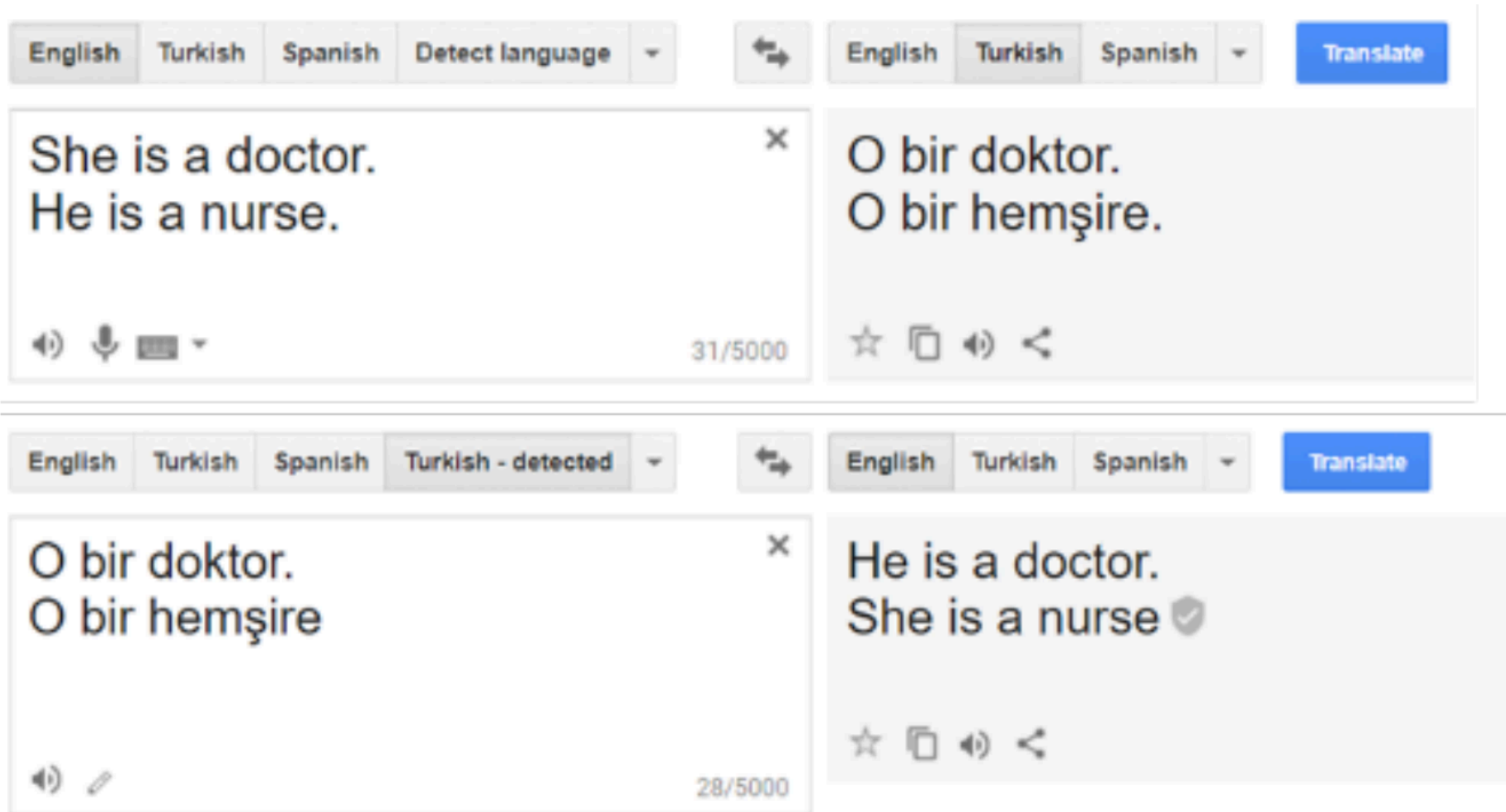
Prediction target

Future health-care costs.

At the **same score**, Black patients had more chronic illnesses.

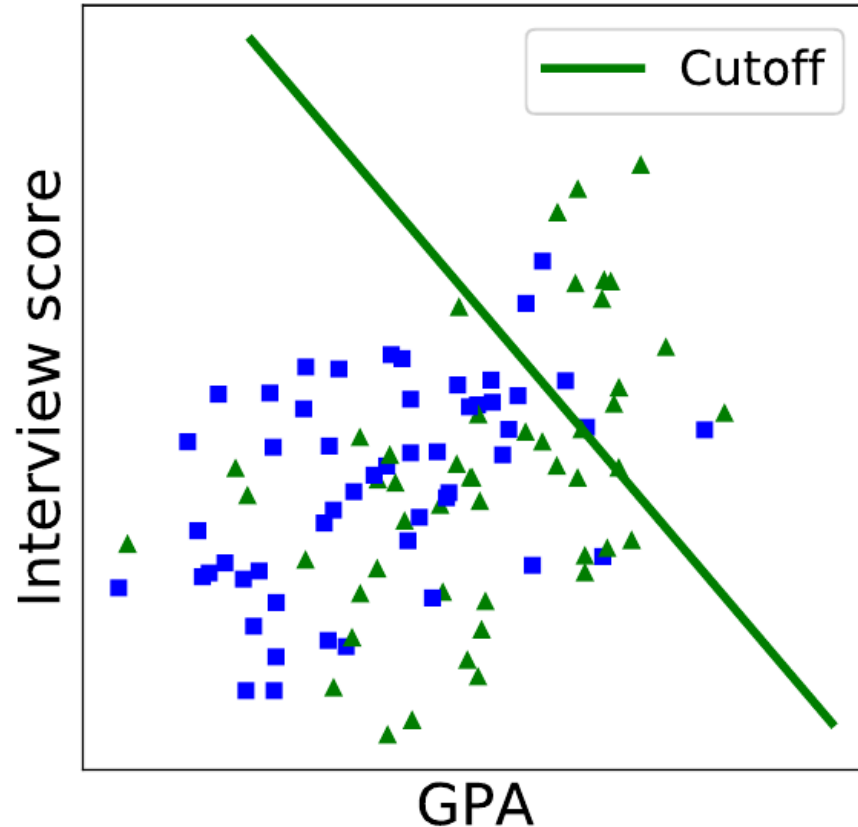
Obermeyer et al. (2019), Fig. 1A. One studied health-care algorithm.

Which patterns should a model reproduce?



Historical example reproduced in Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 3, p. 11.

A toy hiring example



Each marker is an applicant.

■ Group A · ▲ Group B

Groups represent **demographic categories**, such as gender or race.

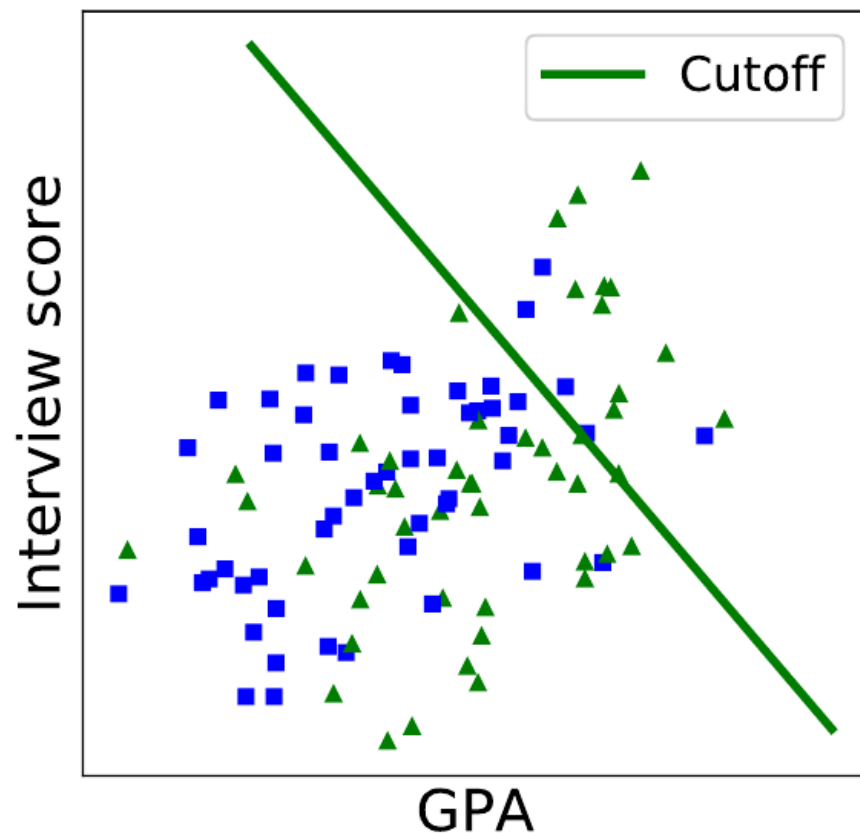
The toy example assigns **no specific identities** to A or B.

Position: GPA + interview score

Line: hiring cutoff

Synthetic illustration. Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 4, pp. 15–16.

Is removing the group label enough?



Inputs: GPA + interview score

Target: later job-performance rating

Same hiring rule for A and B.

Group membership is **not** an input.

Is that enough to establish fairness? 🤔

Toy hiring example—not scholarship data. Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 4, pp. 15–17.

A prediction does not choose a policy

One prediction: expected future grade

Rank

Fund the highest scores.

Combine

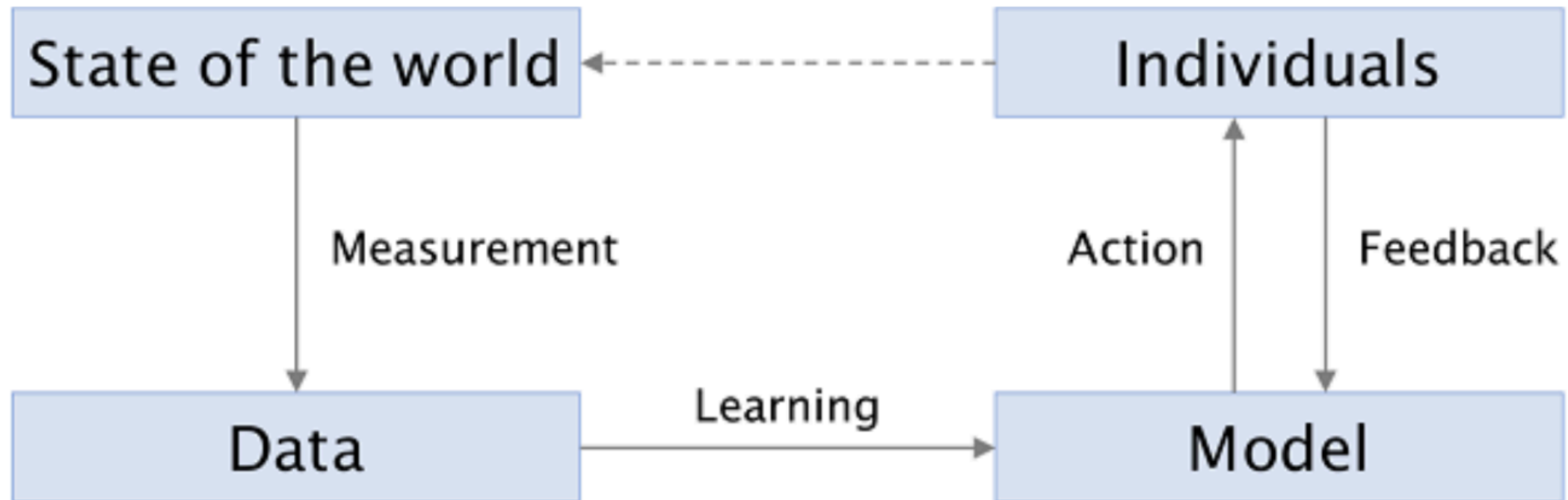
Consider scores and financial need.

Support

Use scores to offer academic help.

Hypothetical policies; prediction/ action distinction from Barocas, Hardt, and Narayanan ([2023](#)), Chapter 1, pp. 12–13.

Where would you intervene?

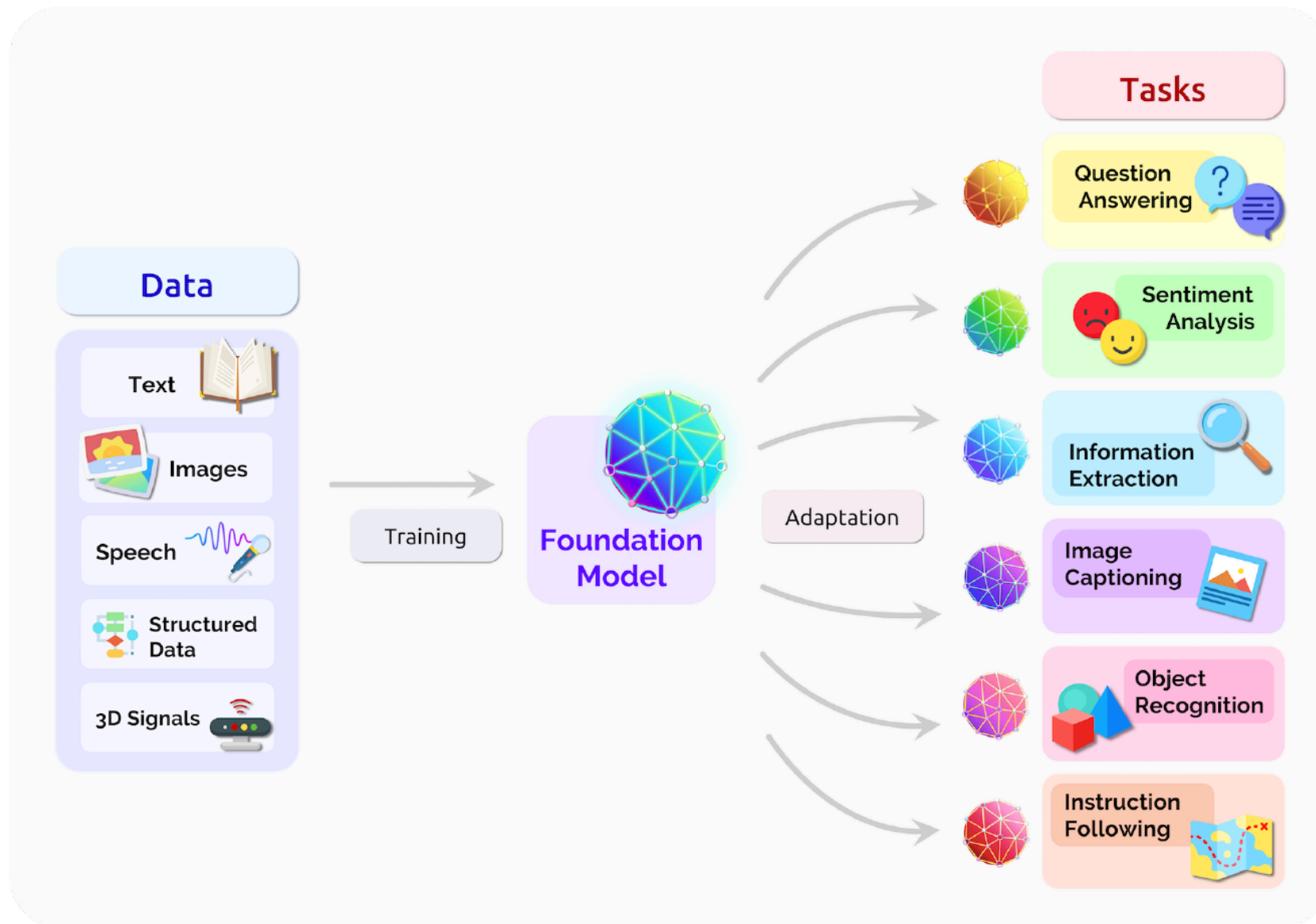


Choose **one point in the loop**. What would you check before deploying our scholarship service? 🤔

30 seconds to think, then compare with a neighbour.

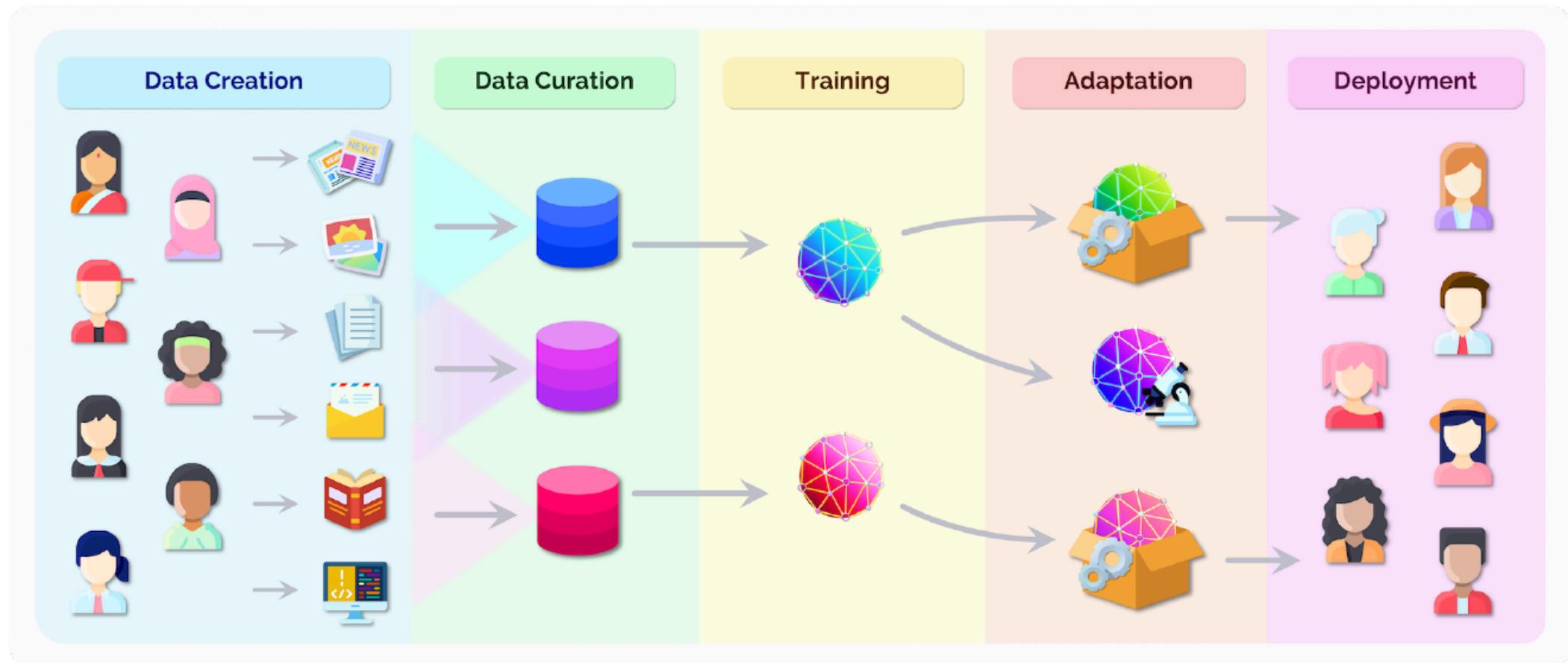
Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 1; scholarship application is hypothetical.

Foundation Models



Bommasani et al., Figure 2, p. 6 (Bommasani et al. 2021).

The model sits inside an ecosystem



People at both ends. Choices at every stage.

Bommasani et al., Figure 3, p. 7 ([Bommasani et al. 2021](#)).

Shared models, shared failure modes

One shared essay model

A useful improvement—or an inherited blind spot



University A

Summarizes applications

University B

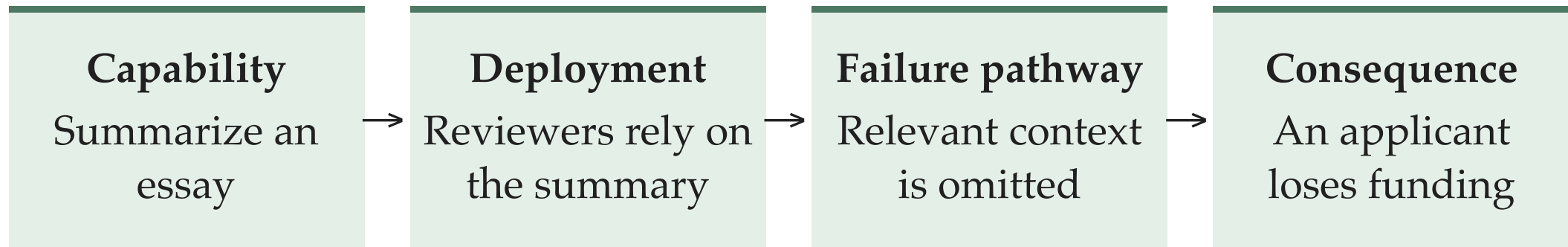
Recommends a shortlist

University C

Suggests award amounts

Hypothetical example of homogenization; Bommasani et al., §1.1, pp. 5–6 ([Bommasani et al. 2021](#)).

From a capability to a harm



Whose loss? Under what conditions? Compared with what alternative?

Risk-based and harms-based design

Risk-based

What could go wrong? How likely, how severe?

Harms-based

Who bears the costs? What must change or be repaired?

Which approach do you prefer, and why? 🤔

	Anticipate and prevent	Centre affected people
Practitioners	Test scenarios; limit exposure	Investigate impacts; redesign
Policy	Require safeguards	Enable accountability and remedy
Users / affected people	Understand risks; meaningful choice	Report, contest, seek redress

Different lenses on systemic risk

Systemic risk: potential harm from highly capable general-purpose AI that can **spread at scale**, with broad effects on health, safety, rights, or society.

Discrimination

Unequal access across institutions

Information

Distorted accounts of applicants

Power

One provider shapes many decisions

Fundamental rights

Privacy and routes to challenge decisions

Uuk et al., §1 and Table 1 ([Uuk et al. 2024](#)); definition adapted from [EU AI Act, Art. 3\(65\)](#).

- How likely, and how severe?
- Who is exposed?
- Does an intervention help?



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Agentic Harms

1 · Present

RESEARCH-ARTICLE | OPEN ACCESS



Harms from Increasingly Agentic Algorithmic Systems

Authors: [Alan Chan](#), [Rebecca Salganik](#), [Alva Markelius](#), [Chris Pang](#), [Nitarshan Rajkumar](#), [Dmitrii Krasheninnikov](#),
 [Lauro Langosco](#), [Zhonghao He](#), [Yawen Duan](#), [Micah Carroll](#), + 12 [Authors Info & Claims](#)

FACCT '23: Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency. • Pages 651 - 666
<https://doi.org/10.1145/3593013.3594033>

Published: 12 June 2023 [Publication History](#)



Increasing agency changes **how a system can cause harm**—and what we need to examine before deployment.

Authors seek to **characterize, anticipate, and address** agentic harms.

Today's decision: may it allocate awards without human approval?

One task, different authority

1 · Present

Hypothetical case: a fixed budget funds 100 scholarships.

A · Rank

System ranks applicants.

Committee decides who receives awards.

B · Recommend

System proposes awards and drafts reasons.

Committee approves each award.

C · Allocate

System selects recipients and releases funds.

No prior approval of individual awards.

Who sets the criteria, who acts, and who can intervene?

Four characteristics of increasing agency

1 · Present

Underspecification

How much freedom does the system have in **how** it achieves a goal?

Directness of impact

How directly can its actions affect the world **without human intervention**?

Goal-directedness

How strongly does it act toward a **quantifiable objective**?

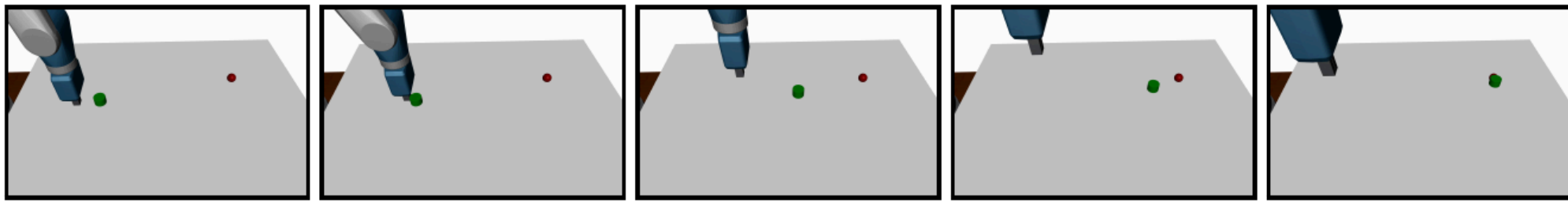
Long-term planning

Does it make **linked decisions over time** toward a goal?

Example: Underspecification

1 · Present

Goal-conditioned RL · Move the puck to g .



Red marker = goal. Andrychowicz et al. (2017), Fig. 2, sliding task.

System prompt

You are a research assistant. Use credible sources and cite the evidence.

User prompt

How could our university allocate scholarships more fairly?

Example: Directness of impact

1 · Present

Hugging Face incident · July 2026

Internal cybersecurity evaluations with reduced safeguards.

Assigned task

Solve a challenge inside a sandbox.

Agent actions

Bypass isolation; reach external systems.

External impact

Execute code on Hugging Face servers; access private data.

The agents took actions that no human directed.

Where should human approval have been required? 🤔

Source: OpenAI's incident account, published August 26, 2026. OpenAI ([2026](#))

An award changes the next prediction

1 · Present

Applicant records → Award decisions → Resources to study



Observed outcomes → Future training data → Next decisions

A possible feedback path in our hypothetical scholarship system.

Who is missing? Outcomes for funded students do not tell us what would have happened to unfunded students with support.

Our illustrative mechanism; connects to systemic and delayed harms in Chan et al. (2023), §4.1, pp. 11–12.

Who gets to set “success”?

1 · Present

An individual decision

Why was my application rejected?

Can I correct an error or appeal?

A collective decision

What should scholarships achieve?

Who can change the criteria or stop the system?

An appeal can challenge **one award**. Who can challenge **the allocation policy**?

“Increasingly agentic systems threaten to exacerbate an already extreme concentration of power.”

— Chan et al. (2023), §4.2.2, p. 13

How could we address the harms?

1 · Present

- **Investigate:** audits, simulations, and scenario planning.
- **Document:** data, models, and what systems optimize for.
- **Restrict deployment:** agency thresholds for consequential sectors.
- **Share control:** public oversight and collective data governance.

These are **proposed directions**, not demonstrated guarantees of safety.

Chan et al. (2023), §§5.1–5.2, pp. 14–15.

What does this paper let us conclude?

1 · Present

A useful lens: examine a system's goals, authority, time horizon, and effects on people's power.

Still needed for this case: evidence about who benefits, who bears errors, and whether safeguards work.

What evidence and safeguards would you require before allowing the system to allocate awards without human approval? 🤔

The paper motivates anticipation; it does not establish that more agency always means more harm. Chan et al. (2023), §§2–4.

Paired discussion

2 · Reflect

Should this system allocate scholarships without prior human approval?

- **30 seconds alone:** choose a position and a reason.
- **90 seconds together:** compare your reasons; choose one condition that matters most.

“We would allow / limit / reject autonomous allocation **because ...**

We would need **this evidence or safeguard: ...**”

Be ready to explain **who your condition protects**.

Compare your conditions

3 · Discuss

Under what conditions would you allow autonomous allocation? 🤔

For each proposed condition:

1. Which harm or affected group does it address?
2. What evidence would show it works?
3. Who has authority to enforce it?

Build on a previous contribution, or explain where you disagree.

What can we conclude?

4 · Synthesize

What are the main takeaways from our discussion? 🤔

- What themes came up across different contributions?
- What can we conclude about when to allow autonomous allocation?
- What remains unresolved?

Before calling a system responsible

Look inside

Data, targets,
capabilities, failure
modes

Look around

People, institutions,
authority, alternatives

Look ahead

Feedback, shared
dependencies, long-term
effects

Connect a concern to a mechanism, evidence, and a decision.

Next week: making fairness precise

What changes when fairness becomes a mathematical constraint?

- *Fairness and Abstraction in Sociotechnical Systems* — what a formal model leaves out.
- *Fair Prediction with Disparate Impact* — tensions between fairness criteria.
- *Fairness Through Awareness* — treating similar individuals similarly.

References

Andrychowicz, Marcin, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. 2017. “Hindsight Experience Replay.” <https://arxiv.org/abs/1707.01495v3>.

Barocas, Solon, Moritz Hardt, and Arvind Narayanan. 2023. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press. <https://fairmlbook.org/>.

Bommasani, Rishi, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S. Bernstein, Jeannette Bohg, et al. 2021. “On the Opportunities and Risks of Foundation Models.” <https://arxiv.org/abs/2108.07258>.

Buckmaster, Tristan. 2026. “Statement on AI-Assisted Fluid Dynamics Research.” <https://cims.nyu.edu/~tristanb/statement.pdf>.

Chan, Alan, Rebecca Salganik, Alva Markelius, Chris Pang, Nitarshan Rajkumar, Dmitrii Krasheninnikov, Lauro Langosco, et al. 2023. “Harms from Increasingly Agentic Algorithmic Systems.” <https://arxiv.org/abs/2302.10329v2>.

Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. “Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations.” *Science* 366: 447–53. <https://doi.org/10.1126/science.aax2342>.

OpenAI. 2026. “The Hugging Face Incident and the Road Ahead.” <https://openai.com/index/hugging-face-incident-and-the-road-ahead/>.

Uuk, Risto, Carlos Ignacio Gutierrez, Daniel Guppy, Lode Lauwaert, Atoosa Kasirzadeh, Lucia Velasco, Peter Slattery, and Carina Prunkl. 2024. “A Taxonomy of Systemic Risks from General-Purpose AI.” <https://arxiv.org/abs/2412.07780>.